

PUJARI NITHISH KUMAR

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PROFILE SUMMARY

Result-oriented Embedded & Software Engineer with a unique blend of low-level system design and modern AI expertise. Proficient in C/C++, RTOS, and ARM Cortex-M, with recent industrial success developing Agentic AI systems and Generative AI solutions using Microsoft Foundry. Currently contributing to a high-scale enterprise platform handling over \$80 billion in circulation by optimizing autonomous agents and prompt engineering. Passionate about combining technical depth in embedded innovations with advanced software capabilities to deliver robust, efficient solutions.

EDUCATION

Skill Lync – NASSCOM, Bengaluru

Masters Certification in Embedded Systems - 71.7%

Apr 2023 – Mar 2024

Bengaluru, Karnataka

GMR Institute of Technology, Rajam

B.Tech – Electrical and Electronics Engineering - 63.8%

2018 - 2022

Rajam, Srikakulam, Andhra Pradesh

Sri Chaitanya Junior College, Srikakulam

Class 12th – MPC - 80.3%

2016 - 2018

Srikakulam, Andhra Pradesh

Visakha High School, Srikakulam

Class 10th – SSC - 92%

2015 - 2016

Srikakulam, Andhra Pradesh

PROFESSIONAL WORK EXPERIENCE

Accenture Solution Private Limited

Apr 2024 – Present

Packaged App Development Associate

Bengaluru, Karnataka

- AI Agent Development: Designing and deploying autonomous Agentic AI systems using Microsoft Foundry (formerly Azure AI Studio), automating complex workflows for a high-scale platform circulating over \$80 billion annually.
- Generative AI & Data Retrieval: Implementing Generative AI solutions integrated with Vector Databases to enable semantic search and efficient context management for large-scale datasets.
- Prompt Engineering: Applying advanced Prompt Engineering techniques to optimize Large Language Model (LLM) performance, ensuring high accuracy, reliability, and safety in critical business operations.
- System Validation: Conducting system-level validation and managing the full lifecycle of software defects using Azure DevOps, ensuring precise documentation and handoff to development leads.
- Technologies Used: Generative AI, Agentic AI, Microsoft Foundry, Prompt Engineering, C#, .NET, Vector DB, Azure DevOps, Agile Methodology, Microsoft Excel, Word, PowerPoint, M365 Copilot.

INDUSTRIAL WORK EXPERIENCE(Internships)

Skill Lync, Bengaluru

Apr 2023 – Mar 2024

Embedded Software Developer Trainee

Bengaluru, Karnataka

- Embedded System Specialisation: Focuses on Embedded systems, Micro Controller architecture, RTOS, Code repository, compliance to ISO26262.
- Embedded System Programming: Emphasises ARM Cortex-M architecture, GPIO, timers, interrupts, BareMetal programming, Linux driver development, and system programming using C.
- Acquired Collaborative hardware design and schematic reading abilities.
- Technologies Used: Embedded systems, GIT, system programming, real-time operating systems, C/C++

Wingfotech Private Limited – Aakar IIT Bombay, Hyderabad

May 2019

Summer Embedded Software Intern

Hyderabad, Telangana

- Acquired experience working on projects utilizing embedded C and robotics.
- Developed a comprehensive understanding of designing automation systems using these technologies.
- Gained experience with platforms such as Blink and Google Assistant to control devices.
- Developed robust problem-solving and communication abilities.
- Technologies Used: C, IFTTT, ATmega328p, Integration.

PROJECTS

High-Scale Enterprise Agentic AI Platform

- Project Goal: Automate complex data retrieval and operational workflows for a platform circulating over \$80 Billion annually.
- System Architecture: Developed autonomous Agentic AI systems using Microsoft Foundry, integrating Vector Databases to enable high-speed, context-aware semantic search across vast datasets.
- Implementation: Utilized Prompt Engineering techniques to optimize Large Language Model (LLM) responses, ensuring strict accuracy and reduced latency in real-time processing.
- Impact: Streamlined critical business operations by deploying intelligent agents capable of handling sophisticated user queries with minimal human intervention.
- Technologies Used: Microsoft Foundry, Generative AI, Vector DB, Agentic AI, Python, Azure DevOps.

Robotics System with Wireless Communication and Path Navigation Using ARM Controller

- System Control: Utilises an STM32g070rb ARM controller for central processing, Bluetooth for wireless control, and an RFID module for secure identification.
- Navigation and Sensing: Employs IR and ultrasonic sensors for precise path navigation and obstacle detection with various sensors/actuators.
- Motion and Actuation: Features a servo motor and motor driver for smooth and precise movement.
- Project Features: Bluetooth control, obstacle avoidance, line following, distance-based navigation, and secure operations through RFID.
- Technologies Used: STM32, HAL, UART, Embedded C.

Auto temperature detection entrance for a COVID safety room

- Project Goal: Develop a system for automatically detecting individuals temperature to improve COVID safety.
- System Function: Use temperature sensors to measure body temperature at the entrance, granting access to individuals with safe temperatures and denying access to those with unsafe temperatures.
- Project Purpose: Provide a convenient and effective way to ensure health and safety in a COVID-safe environment.
- Technologies Used: Arduino UNO, C (Programming Language).

TECHNICAL SKILLS

Coding Languages & Hardware: Generative AI, Agentic AI, Prompt Engineering, Embedded C, Python, BareMetal, FreeRTOS, STM32F/G.

Software's: STM32Cube IDE, Microchip Studio, Visual Studio Code, Putty, Arduino IDE, MS Office, GitHub.

Operating Systems: Windows, MacOS, Ubuntu.

Protocols & Others: UART, I2C, SPI, SDLC, HAL, Git, Kernel Drivers, Device Drivers, GDB Debugging.

SOFT SKILLS

Collaboration, Communication, Leadership, Teamwork, Critical Thinking, Adaptability, Analytical Skills.

ACHIEVEMENTS

- **Certification** - Google Cloud Certified - Professional Machine Learning Engineer 
- Graduated from the **Tech Expressway program at Accenture** with a **87%** passing score in the hands-on C++ assessment. Completed stream training in **C++ and UNIX** - 2024.
- Learned and certified in **Learning Arduino: Foundations, Learning Arduino: Interfacing with Analog Devices** and **Learning PCB design with EAGLE** in LinkedIn Learning – Nov 2020.
- Certified in **Accelerated Career-Program in Embedded Systems (On-Campus) Courseware Partner: IT-ITes SSC NASSCOM** with a **71.7%** passing score in the assessment conducted by NASSCOM at **Skill-Lync** - 2024.
- Participated in a spotlight event **Drone voyage** conducted by STEP CONE at GMRIT and got 3rd Place.
- Participated in a training course on **Interfacing and Programming of Arduino** conducted by ISTE and IETE student chapters at GMRIT-2019.